

# Wasiur R. KhudaBukhsh

*Assistant Professor, School of Mathematical Sciences, University of Nottingham*

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## APPOINTMENTS

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| Oct 2021 – present  | <b>Assistant Professor</b><br>School of Mathematical Sciences, University of Nottingham, UK                                                                     |
| Jan 2020 – Sep 2021 | <b>President's Postdoctoral Scholar</b><br><b>Mathematical Biosciences Institute</b> and College of Public Health, The Ohio State University, Columbus, OH, USA |
| Oct 2018 – Dec 2019 | <b>Postdoctoral Researcher</b><br><b>Mathematical Biosciences Institute</b> and College of Public Health, The Ohio State University, Columbus, OH, USA          |
| Jan 2014 – Sep 2018 | <b>Research Associate</b><br>Department of Electrical Engineering and Information Technology, Technische Universität Darmstadt, Germany                         |
| Jun 2011 – Dec 2013 | <b>Manager, Advanced Analytics</b><br>Business Intelligence Unit, ICICI Bank, Mumbai, India                                                                     |

## EDUCATION

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| 2023 – 2024 | <b>Postgraduate Certificate in Higher Education (PGCHE)</b><br>University of Nottingham, UK                                                                                                                                                                                                           |
| 2023        | <b>Fellow of the Higher Education Academy (FHEA)</b><br>Advance HE, UK                                                                                                                                                                                                                                |
| 2014 – 2018 | <b>PhD (Dr. rer. nat.), summa cum laude</b><br>Department of Electrical Engineering and Information Technology, Technische Universität Darmstadt, Germany<br><i>Thesis: Model reductions for queueing and agent-based systems with applications in communication networks.</i> Advisor: Heinz Koeppel |
| 2009 – 2011 | <b>Master of Statistics (M.Stat.)</b><br><b>Indian Statistical Institute</b> , Kolkata, India                                                                                                                                                                                                         |
| 2006 – 2009 | <b>BSc (Honours) in Statistics</b><br><b>University of Calcutta</b> , Kolkata, India                                                                                                                                                                                                                  |

## RESEARCH INTERESTS

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Probability theory and stochastic processes; mathematical biology.

## GRANTS AND AWARDS

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### *Research funding*

2024 – 2025      Engineering and Physical Sciences Research Council (EPSRC), EP/Y027795/1, principal investigator, £79,987.58 (March 2024 – February 2025)

### *Smaller grants and travel awards*

- Isaac Newton Institute (INI) Retreat, 29 June – 3 July 2026 (with Olga Izyumtseva and Ercan Sönmez)
- Center for Mathematical Machine Learning and its Applications (CM2LA) Seorabol Research Station Workshop grant, South Korea, June 2025 (US\$7,000)
- Isaac Newton Institute (INI) Retreat, 29 July – 4 August 2024
- London Mathematical Society (LMS) Research in Pairs grant, 2024 (£1,200)
- MATRIX–Simons Travel Grant, February 2024 (AU\$2,500)
- University of Nottingham Faculty of Science International Research Collaboration Fund (£3,150)
- Heilbronn Institute for Mathematical Research (HIMR) Small Grants Scheme (£1,000)
- Royal Society International Exchange grant, 2022 (£11,702.09)
- University of Nottingham Faculty of Science International Collaboration Fund (£4,375)
- LMS Undergraduate Research Bursary, 2022 (£900, matched by £900 from the University of Nottingham)
- LMS Research in Pairs grant, 2022 (£1,200)
- Institute of Mathematical Statistics New Researchers Conference, 2022 (US\$1,860)
- American Institute of Mathematics (AIM) SQuaREs (Structured Quartet Research Ensembles), with Hye-Won Kang, Lea Popovic, Grzegorz A. Rempala, Ruth J. Williams, and Felipe Campos
- German Academic Exchange Service (DAAD) travel grant to attend the Annual Meeting of the SMB, Sydney, Australia, 2018 (c. €2,000)

### *Awards*

2024      Best Paper Award, International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)

2019      President's Postdoctoral Scholars Program (PPSP), The Ohio State University

2018      Best-in-Session Presentation Award, IEEE International Conference on Computer Communications (INFOCOM), Honolulu, HI, USA

## PUBLICATIONS

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In mathematics, authors are conventionally listed alphabetically.

### *Preprints and submitted papers*

1. Kushankur Dutta, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempala. Coloured epidemic models: functional law of large numbers and propagation of chaos. 2026.
2. Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, M. Gabriela M. Gomes, and Grzegorz A. Rempala. From individual-based stochastic epidemics to heterogeneous SIR equations. 2026. Submitted.
3. Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Shrinkage priors for Bayesian substitute confounders. 2026.

4. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Fluctuation analysis of the standard QSSA for a stochastic multistage Michaelis–Menten model. 2026. Submitted.
5. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Self-intersection local times for Volterra Gaussian processes in stochastic flows with interaction. 2026. Submitted.
6. J. D. Harborne, **Wasiur R. KhudaBukhsh**, and John R. King. Deterministic and stochastic asymptotics for a multiscale epidemic model. 2026. Revision requested.
7. Yordan P. Raykov, Hengrui Luo, Justin D. Strait, and **Wasiur R. KhudaBukhsh**. Kernel-based estimators for functional causal effects. 2026.
8. **Wasiur R. KhudaBukhsh** and Yangrui Xiang. Mixing time for a noisy SIS model on graphs. 2026. Submitted.
9. Karim S. Elsayed, Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Amr Rizk. Stochastic analysis of entanglement-assisted quantum communication channels. 2025. Revision requested.
10. Olga Izyumtseva and **Wasiur R. KhudaBukhsh**. Local times of self-intersection and sample path properties of Volterra Gaussian processes. 2025. Revision requested.

#### *Peer-reviewed journal articles*

11. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Statistical inference for a multiscale stochastic model of enzyme kinetics via propagation of chaos. *Stochastic Processes and their Applications*, 2027. Accepted for publication.
12. Arnab Ganguly and **Wasiur R. KhudaBukhsh**. Asymptotic analysis of the total quasi-steady state approximation for the Michaelis–Menten enzyme kinetic reactions. *Journal of Mathematical Analysis and Applications*, 561(1):130551, 2026.
13. Yi Fu, Hye-Won Kang, **Wasiur R. KhudaBukhsh**, Lea Popovic, Grzegorz A. Rempała, and Ruth J. Williams. Fragility in a Togashi–Kaneko stochastic model with mutations. *SIAM Journal on Life Sciences*, 1(2):202–228, 2026.
14. Riccardo Corradin, Luca Danese, **Wasiur R. KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of time-dependent observations with common structural changes. *Statistics and Computing*, 36(7), 2026.
15. Alexander E. Zarebski, Nefel Tellioglu, Jessica E. Stockdale, Julie A. Spencer, **Wasiur R. KhudaBukhsh**, Joel C. Miller, and Cameron Zachreson. Including frameworks of public health ethics in computational modelling of infectious disease interventions. *Interface Focus*, 15, 2025.
16. Yushuf Sharker, Zaynab Diallo, **Wasiur R. KhudaBukhsh**, and Eben Kenah. Pairwise accelerated failure time regression models for infectious disease transmission in close contact groups with external sources of infection. *Statistics in Medicine*, 2024.
17. **Wasiur R. KhudaBukhsh** and Grzegorz A. Rempała. How to correctly fit an SIR model to data from an SEIR model? *Mathematical Biosciences*, 2024.
18. Matthew Wascher, Patrick Schnell, **Wasiur R. KhudaBukhsh**, Mikkel B.M. Quam, Joseph Tien, and Grzegorz Rempała. Estimating disease transmission in a closed population under repeated testing. *Journal of the Royal Statistical Society: Series C (JRSSC)*, 2024.
19. István Z. Kiss, Luc Berthouze, and **Wasiur R. KhudaBukhsh**. Towards inferring network properties from epidemic data. *Bulletin of Mathematical Biology*, 2024.
20. **Wasiur R. KhudaBukhsh**, Sat Kartar Khalsa, Eben Kenah, Grzegorz Rempała, and Joseph Tien. COVID-19 dynamics in an Ohio prison. *Frontiers in Public Health*, 2023.
21. **Wasiur R. KhudaBukhsh**, Caleb Deen Bastian, Matthew Wascher, Colin Klaus, Saumya Yashmohini Sahai, Mark H. Weir, Eben Kenah, Elisabeth Root, Joseph H. Tien, and Grzegorz A. Rempała. Projecting COVID-19 cases and hospital burden in Ohio. *Journal of Theoretical Biology*, 561:111404, 2023.

22. Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, and Grzegorz Rempała. Likelihood-free dynamical survival analysis applied to the COVID-19 epidemic in Ohio. *Mathematical Biosciences and Engineering*, 20, 2023.
23. Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koepl. Hypergraphon mean-field games. *Chaos*, 2022.
24. **Wasiur R. KhudaBukhsh**, Casper Woroszylo, Grzegorz Rempała, and Heinz Koepl. A functional central limit theorem for SI processes on configuration model graphs. *Advances in Applied Probability*, 2022.
25. Colin Klaus, Matthew Wascher, **Wasiur R. KhudaBukhsh**, Joseph H. Tien, Grzegorz A. Rempała, and Eben Kenah. Assortative mixing among vaccination groups and biased estimation of reproduction numbers. *The Lancet Infectious Diseases*, 22:P579–581, 2022.
26. Francesco Di Lauro\*, **Wasiur R. KhudaBukhsh\***, István Z. Kiss, Eben Kenah, Max Jensen, and Grzegorz Rempała. Dynamic survival analysis for non-Markovian epidemic models. *Journal of the Royal Society Interface*, 2022. \*Joint first authors.
27. Kai Cui, **Wasiur R. KhudaBukhsh**, and Heinz Koepl. Motif-based mean-field approximation of interacting particles on clustered networks. *Physical Review E*, 105, 2022.
28. Harley Vossler, Pierre Akilimali, Yuhan Pan, **Wasiur R. KhudaBukhsh**, Eben Kenah, and Grzegorz A. Rempała. Analysis of individual-level epidemic data: study of 2018–2020 Ebola outbreak in Democratic Republic of the Congo. *Scientific Reports*, 12, 2022.
29. Ido Somekh\*, **Wasiur R. KhudaBukhsh\***, Elisabeth Dowling Root\*, Greg Rempala, Eric Simoes, and Eli Somekh. Quantifying the population-level effect of COVID-19 mass vaccination campaign in Israel: a modeling study. *Open Forum Infectious Diseases*, 2022. \*Joint first authors.
30. **Wasiur R. KhudaBukhsh**, Hye-Won Kang, Eben Kenah, and Grzegorz Rempała. Incorporating age and delay into models for biophysical systems. *Physical Biology*, 18(1), 2021. Invited paper.
31. **Wasiur R. KhudaBukhsh**, Boseung Choi, Eben Kenah, and Grzegorz Rempała. Survival dynamical systems: individual-level survival analysis from population-level epidemic models. *Journal of the Royal Society Interface Focus*, 10(1), 2020.
32. **Wasiur R. KhudaBukhsh**, Arnab Auddy, Yann Disser, and Heinz Koepl. Approximate lumpability for Markovian agent-based models using local symmetries. *Journal of Applied Probability*, 56, 2019.
33. Hye-Won Kang\*, **Wasiur R. KhudaBukhsh\***, Heinz Koepl, and Grzegorz Rempała. Quasi-steady-state approximations derived from a stochastic enzyme kinetics. *Bulletin of Mathematical Biology*, 81(5):1303–1336, 2019. \*Joint first authors.
34. Saumya Yashmohini Sahai, Saket Gurukar, **Wasiur R. KhudaBukhsh**, Srinivasan Parthasarathy, and Grzegorz A. Rempała. A machine learning model for nowcasting epidemic incidence. *Mathematical Biosciences*, 2021.
35. **Wasiur R. KhudaBukhsh**, Sounak Kar, Bastian Alt, Amr Rizk, and Heinz Koepl. Generalized cost-based job scheduling in very large heterogeneous cluster systems. *IEEE Transactions on Parallel and Distributed Systems*, 31(11):2594–2604, 2020.
36. Boseung Choi, Sydney Busch, Dieudonné Kazadi, Benoit Ilunga, Emile Okitolonda, Yi Dai, Robert Lumpkin, Omar Saucedo, **Wasiur R. KhudaBukhsh**, Joseph Tien, Marcel Yotebieng, Eben Kenah, and Grzegorz A. Rempała. Modeling outbreak data: analysis of a 2012 Ebola virus disease epidemic in DRC. *BIOMATH*, 8(2), 2019.
37. **Wasiur R. KhudaBukhsh**, Amr Rizk, Sounak Kar, and Heinz Koepl. Provisioning and performance evaluation of parallel systems with output synchronization. *ACM Transactions on Modeling and Performance Evaluation of Computing Systems (TOMPECS)*, 4(1), 2019.
38. Bastian Alt, Markus Weckesser, Christian Becker, Matthias Hollick, Sounak Kar, Anja Klein,

Robin Klose, Roland Kluge, Heinz Koepl, Boris Koldehofe, **Wasiur R. KhudaBukhsh**, Mahdi Mousavi, Martin Pfannemueller, Amr Rizk, Andy Schuerr, and Ralf Steinmetz. Transitions: a protocol-independent view of the future internet. *Proceedings of the IEEE*, 107(4):835–846, 2019.

#### *Book chapters*

39. Olga Izyumtseva, **Wasiur R. KhudaBukhsh**, and Grzegorz A. Rempała. Functional law of large numbers for an epidemic model with random effects. *Handbook of Statistics*. Elsevier, 2024.
40. Grzegorz A. Rempała and **Wasiur R. KhudaBukhsh**. Dynamical survival analysis for epidemic modeling. In *Handbook of Visual, Experimental and Computational Mathematics*, pages 1–17. Springer International Publishing, 2023.

#### *Peer-reviewed conference proceedings*

41. Karim Elsayed, **Wasiur R. KhudaBukhsh**, and Amr Rizk. On the trade-off between fidelity and latency for the quantum link layer with few memories and entanglement purification. In *Proceedings of the International Conference on Quantum Communications, Networking, and Computing (QCNC 2024)*, 2024. Best Paper Award.
42. Karim Elsayed, **Wasiur R. KhudaBukhsh**, and Amr Rizk. On the fidelity distribution of link-level entanglements under purification. In *Proceedings of the IEEE International Conference on Communication (ICC) 2024*, 2024.
43. Riccardo Corradin, Luca Danese, **Wasiur R. KhudaBukhsh**, and Andrea Ongaro. Model-based clustering of non-stationary time series with common historical change times. In *Statistical Learning, Sustainability and Impact*, 2023.
44. **Wasiur R. KhudaBukhsh**, Bastian Alt, Sounak Kar, Amr Rizk, and Heinz Koepl. Collaborative uploading in heterogeneous networks: optimal and adaptive strategies. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2018. <20% acceptance rate. Best-in-Session Presentation Award.
45. **Wasiur R. KhudaBukhsh**, Amr Rizk, Alexander Frömmgen, and Heinz Koepl. Optimizing stochastic scheduling in fork-join queueing models: bounds and applications. In *IEEE International Conference on Computer Communications (INFOCOM)*, 2017. ~20% acceptance rate.
46. Adrian Šošić, **Wasiur R. KhudaBukhsh**, A. M. Zourbir, and Heinz Koepl. Inverse reinforcement learning in swarm systems. In *AAMAS Workshop on Transfer in Reinforcement Learning*, 2017. Available: <http://www.tirl.info/proceedings/2017/SosicEtal-Inverse.pdf>.
47. Adrian Šošić, **Wasiur R. KhudaBukhsh**, A. M. Zourbir, and Heinz Koepl. Inverse reinforcement learning in swarm systems. In *International Conference on Autonomous Agents & Multiagent Systems (AAMAS)*, 2017. ~26% acceptance rate. Best Paper Award finalist.
48. **Wasiur R. KhudaBukhsh**, Julius Rückert, Julian Wulfheide, David Hausheer, and Heinz Koepl. Analysing and leveraging client heterogeneity in swarming-based live streaming. In *IFIP Networking Conference (IFIP Networking) and Workshops*, 2016. ~26% acceptance rate.
49. Mahdi Mousavi, Hussein Al Shatri, **Wasiur R. KhudaBukhsh**, Heinz Koepl, and Anja Klein. Cross-layer QoE-based incentive mechanism for video streaming in multi-hop wireless networks. In *IEEE 86th Vehicular Technology Conference (VTC)*, 2017.

## Thesis and technical notes

50. **Wasiur R. KhudaBukhsh**. *Model reductions for queueing and agent-based systems with applications in communication networks*. PhD thesis, Technische Universität, Darmstadt, 2018. Available at: <http://tuprints.ulb.tu-darmstadt.de/7588/>.
51. Mark Sinzger-D'Angelo, Heinz Koepl, and **Wasiur R. KhudaBukhsh**. Bounds on the spectral radius of real-valued non-negative kernels on measurable spaces. 2023. Preprint: <https://arxiv.org/abs/1808.00258>.

## TALKS

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60 invited and 30 contributed or local talks; 7 poster presentations.

### Invited talks

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|----------|-----------------------------------------------------------------------------------------------------------------------------|
| Jun 2026 | Stochastic Processes and their Applications (SPA 2026), Ithaca, NY, USA                                                     |
| May 2026 | Workshop “Individual variation and its consequences for population dynamics”, University of Strathclyde, Glasgow, UK        |
| Apr 2026 | Probability Seminar, Louisiana State University, USA                                                                        |
| Mar 2026 | Mathematics in Life Sciences Seminar, University of Strathclyde, UK                                                         |
| Jan 2026 | Department Seminar, School of Mathematical and Statistical Sciences, IIT Mandi, India                                       |
| Dec 2025 | International Conference on Recent Advances in Medical Science and Technology, IIT Kharagpur, India                         |
| Nov 2025 | HEALMOD Seminar, The Ohio State University, USA (online)                                                                    |
| Oct 2025 | Statistics and Probability Seminar, University of Manchester, UK                                                            |
| Sep 2025 | Workshop on Gaussian and Lévy-type stochastic processes, University of Nottingham, UK                                       |
| Sep 2025 | Annual Meeting of the Royal Statistical Society, Edinburgh, UK                                                              |
| Jul 2025 | Stochastic Processes and their Applications (SPA 2025), Wrocław, Poland                                                     |
| Jun 2025 | MIDAS Webinar (online)                                                                                                      |
| Jun 2025 | BIRS-CMO Workshop “Mathematical and statistical challenges in post-pandemic epidemiology and public health”, Oaxaca, Mexico |
| May 2025 | Statistics and Probability Seminar, University of Essex, UK                                                                 |
| Dec 2024 | Innovations in Machine Learning, Artificial Intelligence, Data Science and Modelling (IMLAIDSM'24), India                   |
| Nov 2024 | Probability Seminar, Louisiana State University, USA                                                                        |
| Nov 2024 | Mathematics of Reaction Networks (MoRN) Seminar                                                                             |
| Oct 2024 | CMMB Seminar, University of Nottingham, UK                                                                                  |
| Jun 2024 | Mini-symposium, Annual Meeting of the Society for Mathematical Biology, Seoul, South Korea                                  |
| Jun 2024 | Epidemiology Workshop, Konkuk University, Seoul, South Korea                                                                |
| Dec 2023 | Mathematical Biology Seminar, University of Leeds, UK                                                                       |
| Dec 2023 | Mathematical Life Sciences talk, University of Bonn, Germany                                                                |
| Nov 2023 | Probability Seminar, Louisiana State University, USA                                                                        |
| Aug 2023 | Mini-symposium, International Congress on Industrial and Applied Mathematics (ICIAM), Tokyo, Japan                          |

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| Jul 2023 | SMB Diversity in Math Bio seminar series                                                                               |
| Jun 2023 | Mini-symposium, SIAM Conference on Optimization, Seattle, WA, USA                                                      |
| Apr 2023 | BBSRC DTP event, University of Nottingham, UK                                                                          |
| Apr 2023 | Probability Seminar, Louisiana State University, USA                                                                   |
| Mar 2023 | Stochastics Seminar, Georgia Tech, USA (online)                                                                        |
| Feb 2023 | NANT Lecture, Networking in Applied Network Theory, Będlewo, Poland (online)                                           |
| Dec 2022 | Theoretical Statistics and Mathematics Seminar, Indian Statistical Institute, New Delhi, India                         |
| Dec 2022 | Workshop on Non-equilibrium Phenomena in Physics and Biology, Asia Pacific Center for Theoretical Physics, South Korea |
| Dec 2022 | Self-Organizing Systems Workshop, Technische Universität Darmstadt, Germany                                            |
| Apr 2022 | Mini-symposium, SIAM Conference on Uncertainty Quantification                                                          |
| Feb 2022 | Statistics and Probability Seminar, University of Nottingham, UK                                                       |
| Jan 2022 | Mathematics of Reaction Networks (MoRN) Seminar                                                                        |
| Nov 2021 | Math Bio Seminar, The Ohio State University, USA (online)                                                              |
| Oct 2021 | CMMB Seminar, University of Nottingham, UK                                                                             |
| Oct 2021 | Stochastic Systems Seminar, Department of Mathematics, University of California San Diego, USA (online)                |
| Sep 2021 | Guest lecture on infectious disease epidemiology, The Ohio State University, USA (online)                              |
| Jul 2021 | American Institute of Mathematics (AIM) workshop “Limits and control of stochastic reaction networks”                  |
| Apr 2021 | Colloquium, Department of Mathematics, Simon Fraser University, Canada (online)                                        |
| Jan 2021 | Special session, Joint Mathematics Meetings (JMM) (online)                                                             |
| Dec 2020 | IDI Virtual COVID-19 Symposium, The Ohio State University, USA (online)                                                |
| Nov 2020 | Seminar, Institute of Applied Mathematics and Mechanics, University of Warsaw, Poland                                  |
| Nov 2020 | Seminar, Institute of Applied Mathematics and Mechanics, University of Warsaw, Poland                                  |
| Oct 2020 | Biostatistics Seminar, The Ohio State University, Columbus, OH, USA                                                    |
| Sep 2020 | Math Bio Seminar, Virginia Tech, USA (online)                                                                          |
| Sep 2020 | International Webinar Series on Artificial Intelligence and Machine Learning (online)                                  |
| Aug 2020 | Workshop “Life on Planet Earth: Above and Below”, Mathematical Biosciences Institute, The Ohio State University, USA   |
| Jul 2020 | Special session on data science, Business Intelligence Unit, ICICI Bank, Mumbai, India                                 |
| Jul 2020 | International Webinar, Departments of Mathematics and Physics, Bidhan Chandra College, India (online)                  |
| May 2020 | Special session, The Erdős Institute Python Bootcamp (with Marissa Renardy)                                            |
| Feb 2020 | Biostatistics Seminar, The Ohio State University, Columbus, OH, USA                                                    |
| Feb 2020 | Mathematical Modeling and Statistical Analysis of Infectious Disease Outbreaks, CIRM, Marseille, France                |

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| 2019     | Mini-symposium, Annual Meeting of the Society for Mathematical Biology, Montreal, Canada                                              |
| Jun 2019 | Mini-symposium, International Conference on Mathematical Methods and Models in Biosciences (BioMath), Będlewo, Poland                 |
| May 2019 | BIRS-CMO Workshop “Scaling Limits of Dynamical Processes on Random Graphs”, Oaxaca, Mexico                                            |
| 2018     | Mini-symposium, Joint Annual Meeting of the SMB and JSMB, Sydney, Australia                                                           |
| May 2017 | Cloud Computing, Machine Learning and Networking Research (CLAN) Lab, Purdue University, USA (host: <a href="#">Vaneet Aggarwal</a> ) |

#### *Contributed talks*

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| Dec 2026 | MIDAS Webinar (online)                                                                                  |
| Jun 2026 | Stochastic Processes and their Applications (SPA 2026), Ithaca, NY, USA                                 |
| Sep 2025 | Annual Meeting of the Royal Statistical Society, Edinburgh, UK                                          |
| Jun 2024 | Melbourne Mathematical Biology Seminar, University of Melbourne, Australia                              |
| Jun 2024 | IEEE International Conference on Communications (ICC), Denver, CO, USA                                  |
| Jul 2023 | Mini-symposium, Annual Meeting of the Society for Mathematical Biology, Columbus, OH, USA               |
| Jul 2023 | Dynamical Systems in the Life Sciences, The Ohio State University, USA                                  |
| Jun 2022 | BIRS Workshop “Preparing for the next pandemic”                                                         |
| Jun 2021 | Annual Meeting of the Society for Mathematical Biology                                                  |
| Jun 2021 | Workshop on Mathematical and Computational Biology                                                      |
| May 2021 | SIAM Conference on Applications of Dynamical Systems (DS21)                                             |
| Mar 2021 | ENAR Spring Meeting                                                                                     |
| Mar 2021 | Mathematical Biosciences Institute (MBI) Seminar, Columbus, OH, USA                                     |
| Feb 2021 | Joint Mathematical Epidemiology and Math Education SMB Subgroup Meeting                                 |
| Nov 2020 | Applied Mathematics Seminar, Department of Mathematics, The Ohio State University, USA                  |
| Aug 2020 | Bernoulli-IMS One World Symposium (online)                                                              |
| Aug 2020 | Mini-symposium, Annual Meeting of the Society for Mathematical Biology (online)                         |
| Aug 2020 | Joint Statistical Meetings (JSM) (online)                                                               |
| Jun 2020 | Mini-symposium, SIAM Conference on Mathematics of Data Science (online)                                 |
| May 2020 | MBI Workshop on Mathematical and Computational Methods in Biology                                       |
| Dec 2019 | International Indian Statistical Association Conference (IISA 2019), Mumbai, India                      |
| Dec 2019 | Latin American Congress of Probability and Mathematical Statistics (CLAPEM XV), Mérida, Yucatán, Mexico |
| Oct 2019 | Mathematical Biosciences Institute (MBI) Seminar, Columbus, OH, USA                                     |
| 2019     | Joint Statistical Meetings (JSM), American Statistical Association, Denver, CO, USA                     |
| Jul 2019 | Stochastic Processes and their Applications, Bernoulli Society, Northwestern University, USA            |

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| Jun 2019 | International Conference on Mathematical Methods and Models in Biosciences (BioMath), Poland |
| 2019     | Mathematical Biosciences Institute (MBI) Seminar, Columbus, OH, USA                          |
| 2018     | IEEE International Conference on Computer Communications (INFOCOM), Honolulu, HI, USA        |
| 2017     | IEEE International Conference on Computer Communications (INFOCOM), Atlanta, GA, USA         |
| 2016     | IFIP Networking Conference and Workshops, Vienna, Austria                                    |

### *Posters*

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| Sep 2022 | Annual Meeting of the Society for Mathematical Biology, Heidelberg, Germany                       |
| Aug 2022 | 22nd Meeting of New Researchers in Statistics and Probability, George Mason University, USA       |
| Feb 2020 | Mathematical Models in Evolutionary Biology, CIRM, Marseille, France                              |
| Sep 2019 | Annual Meeting of the Infectious Diseases Institute, The Ohio State University, Columbus, OH, USA |
| 2019     | Joint Statistical Meetings (JSM), American Statistical Association, Denver, CO, USA               |
| Jun 2019 | CMCF Algorithms and Methods for Single Cell Genomics, University of California, Irvine, USA       |
| May 2019 | BIRS-CMO Workshop “Scaling Limits of Dynamical Processes on Random Graphs”, Oaxaca, Mexico        |

## TEACHING

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### *University of Nottingham, UK (module leader or co-lecturer)*

|             |                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 – 26   | MATH3060 Financial Mathematics stream (with Michael Tretyakov)                                                                                     |
| 2025, 2026  | MATH2109 Probability 3 / Markov Chains (c. 250 and 270 students)                                                                                   |
| 2022 – 2024 | MATH2010 Probability Models and Methods (c. 270, 260 and 282 students)                                                                             |
| 2022 – 25   | MATH4045 Mathematical Finance stream (2022–23 with Fabrizio Leisen; 2023–24 with Fabrizio Leisen and John Billingham; 2024–25 with Marco Iglesias) |

### *The Ohio State University, USA (instructor)*

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|-------------|----------------------------------------------------------------------------------------------------------------------------------|
| Autumn 2019 | STAT 4201 Introduction to Mathematical Statistics, Department of Statistics (over 120 students)                                  |
| Summer 2019 | PUBHBIO 7193 Individual Studies in Biostatistics (advanced stochastic epidemiology), Division of Biostatistics (with Eben Kenah) |

### *Technische Universität Darmstadt, Germany (teaching assistant)*

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|------------|--------------------------------------------------------------------------------------------------------------------------|
| 2016, 2017 | Computational Methods for Systems and Synthetic Biology, Department of Electrical Engineering and Information Technology |
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## SUPERVISION

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Supervised or co-supervised 17 MSc and 16 BSc students (not including REU students).

### *Postdoctoral researchers (University of Nottingham)*

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|-------------|----------------------------------------------|
| 2023 – 2026 | Olga Izyumtseva (British Academy Fellowship) |
| 2024 – 2025 | Yangrui Xiang                                |

### *PhD students (University of Nottingham)*

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|-------------|-----------------------------------------------------------------------------------------------------|
| 2023 – 2027 | Kushankur Dutta (expected April 2027; secondary supervisors: Mikhail Tretyakov and Karthik Bharath) |
| 2023 – 2026 | Wesam El Buaishi (expected December 2026; primary supervisor: Yordan Raykov)                        |
| 2022 – 2026 | James Harborne (viva November 2026; primary supervisor: John King)                                  |

### *MSc students*

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|------|----------------------------------------------------------------------------------------------------------|
| 2026 | Jennifer Wood, MSc dissertation, University of Nottingham                                                |
| 2025 | Wenhao Qiu, MSc dissertation, University of Nottingham                                                   |
| 2025 | Maxwell Hancock, MSc dissertation, University of Nottingham                                              |
| 2025 | William Bateman, MSc dissertation, University of Nottingham                                              |
| 2024 | Siyu Xu, MSc dissertation, University of Nottingham                                                      |
| 2024 | Qi Zhang, MSc dissertation, University of Nottingham                                                     |
| 2024 | Kaustav Chakraborty, MSc dissertation, University of Nottingham                                          |
| 2023 | Nazreen Asok, MSc dissertation, University of Nottingham                                                 |
| 2023 | Changqing Du, MSc dissertation, University of Nottingham                                                 |
| 2022 | Yajie Guo, MSc dissertation, University of Nottingham                                                    |
| 2022 | Jianing Ye, MSc dissertation, University of Nottingham                                                   |
| 2019 | Robert Lumpkin, master's project, The Ohio State University (with Grzegorz A. Rempała; resulted in [36]) |
| 2018 | Ranjani Krishnan, master's thesis, Technische Universität Darmstadt (with Denny Stohr)                   |
| 2018 | Sayantan Kumar, summer internship, Technische Universität Darmstadt                                      |
| 2017 | Arnab Auddy, summer internship, Technische Universität Darmstadt (resulted in [32])                      |
| 2017 | Markus Schanz, master's thesis, Technische Universität Darmstadt (with Christian Koch)                   |
| 2017 | Hameer Abbasi, master's thesis, Technische Universität Darmstadt (with Bastian Alt)                      |

### *BSc students*

|      |                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------|
| 2025 | Yiwen Yan, summer internship, University of Nottingham                                                            |
| 2023 | Maoduan Ran, summer internship, University of Nottingham                                                          |
| 2022 | Sarina Rivlin-Sanders, summer internship, University of Nottingham                                                |
| 2022 | Junzhe Zhu, summer internship, University of Nottingham                                                           |
| 2019 | Jasen Lai, Ran An, Stanley L. Cao, Hongyi Wang, Joey Zeng, and Kai Li, summer projects, The Ohio State University |

|      |                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------|
| 2019 | Yi Dai and Jordan Kirsch, summer projects, The Ohio State University (with Grzegorz A. Rempala and Eben Kenah) |
| 2018 | Siddhaarth Sarkar, summer internship, Technische Universität Darmstadt                                         |
| 2017 | Simon Schwanz, project seminar, Technische Universität Darmstadt (with Christian Koch)                         |
| 2016 | Vikash Vikash, summer internship, Technische Universität Darmstadt                                             |
| 2016 | Simon Schwanz, proseminar, Technische Universität Darmstadt                                                    |

## SERVICE

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### *Editorial*

|                |                                                                                                                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 – present | Editorial Board Member, <i>Mathematical Biosciences</i>                                                                                                                                                                                                |
| 2023           | Guest Associate Editor (on behalf of the Applied Probability Section of the RSS), discussion paper for the <i>Journal of the Royal Statistical Society: Series B</i><br>Guest Editor, special issue of <i>Mathematical Biosciences and Engineering</i> |

### *Committees*

|                |                                                                                                   |
|----------------|---------------------------------------------------------------------------------------------------|
| 2025 – present | Secretary and Treasurer, Reaction Networks Subgroup of the Society for Mathematical Biology (SMB) |
| 2023 – present | Committee member, Applied Probability Section of the Royal Statistical Society (RSS)              |

### *PhD examining*

|      |                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------|
| 2025 | Internal examiner: Liam Critcher, School of Mathematical Sciences, University of Nottingham, UK                       |
| 2025 | External examiner: Dániel Keliger, Institute of Mathematics, Budapest University of Technology and Economics, Hungary |
| 2024 | External examiner: Hiu Ching Yip, Politecnico di Torino, Italy                                                        |

### *Grant review*

|      |                                                                                        |
|------|----------------------------------------------------------------------------------------|
| 2024 | Engineering and Physical Sciences Research Council (EPSRC), UK Research and Innovation |
| 2024 | Netherlands Organisation for Health Research and Development (ZonMw)                   |

### *Organising conferences and workshops*

|          |                                                                                                                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|
| Jul 2026 | Stochastic Reaction Networks subgroup mini-symposium, joint Annual Meeting of the ESMTB and SMB, Graz, Austria (with Stefan Müller) |
| Jul 2026 | Mini-symposium, joint Annual Meeting of the ESMTB and SMB, Graz, Austria (with Olga Izyumtseva and Arnab Ganguly)                   |
| Jun 2026 | Contributed session, Stochastic Processes and their Applications 2026, Cornell University, USA (with Olga Izyumtseva)               |
| May 2026 | Workshop “Recent advances in stochastic processes: theory and applications”, University of Edinburgh, UK (with Tess Vernon)         |
| Sep 2025 | Workshop “Gaussian and Lévy-type stochastic processes”, University of Nottingham, UK (with Olga Izyumtseva)                         |

|          |                                                                                                                                                                                           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jun 2025 | BIRS Workshop “Mathematical and statistical challenges in post-pandemic epidemiology and public health” (with Jessica Stockdale, Sara Del Valle, Joel C. Miller, and Grzegorz A. Rempala) |
| Jul 2024 | Satellite Workshop on Chemical Reaction Networks, Gyeongju, South Korea (with Hye-Won Kang and Jinsu Kim)                                                                                 |
| Jun 2024 | Mini-symposium, Annual Meeting of the Society for Mathematical Biology, Seoul, South Korea (with Hye-Won Kang and Jinsu Kim)                                                              |
| Oct 2023 | Workshop on Epidemic Modelling, Applied Probability Section of the Royal Statistical Society (with Alex Cox, Fraser Daly, and Lucy Teece)                                                 |
| Jul 2023 | Mini-symposium, Annual Meeting of the SMB, Columbus, OH, USA (with Hye-Won Kang)                                                                                                          |
| May 2023 | Mini-symposium, SIAM Conference on Optimization, Seattle, WA, USA                                                                                                                         |
| Sep 2022 | Workshop “Recent challenges in mathematical epidemiology”, University of Nottingham, UK (with Kirsty Bolton and Yordan Raykov)                                                            |
| Sep 2022 | Mini-symposium, European Conference on Mathematical and Theoretical Biology (ECMTB 2022) (with Hye-Won Kang)                                                                              |
| Jun 2022 | BIRS Workshop “Preparing for the next pandemic” (with Sara Del Valle, Joel C. Miller, and Rick Durrett)                                                                                   |
| 2021     | Mini-symposium, SIAM Conference on Applications of Dynamical Systems                                                                                                                      |
| 2021     | Mini-symposium, Annual Meeting of the Society for Mathematical Biology (with Hye-Won Kang)                                                                                                |
| 2021     | Invited session, ENAR Annual Meeting (with Marissa Renardy)                                                                                                                               |
| 2020     | Mini-symposium, Annual Meeting of the Society for Mathematical Biology (with Hye-Won Kang)                                                                                                |
| Jun 2020 | Mini-symposium, SIAM Conference on Mathematics of Data Science (MDS20), Cincinnati, OH, USA (with Veronica Ciocanel)                                                                      |
| Dec 2019 | Contributed session, Latin American Congress of Probability and Mathematical Statistics (CLAPEM XV), Mérida, Yucatán, Mexico (with Hye-Won Kang)                                          |
| Dec 2019 | Session, International Indian Statistical Association Conference (IISA 2019), Mumbai, India (with Arindam Fadikar)                                                                        |

#### *Department and university*

|                |                                                                              |
|----------------|------------------------------------------------------------------------------|
| 2022 – present | Course Director, BSc/MMath with a year in industry, University of Nottingham |
| 2022 – 2023    | Organiser, Statistics and Probability Seminars, University of Nottingham     |
| 2021           | Reviewer, President’s Research Excellence awards, The Ohio State University  |
| 2019           | Judge, OPA travel awards, The Ohio State University                          |
| 2018, 2019     | Judge, OSU Mathematical Contest in Modeling, The Ohio State University       |

#### *COVID-19 modelling response*

Member of the OSU/IDI COVID-19 response modelling team, providing daily predictions of COVID-19 cases and the resulting hospital burden in Ohio to the Ohio Department of Health and the Ohio Hospital Association.

### *Tutorials and REU mentoring*

|      |                                                                                                      |
|------|------------------------------------------------------------------------------------------------------|
| 2020 | “Modelling the COVID-19 pandemic”, Erdős Institute bootcamp                                          |
| 2019 | “How to write scientific papers”, MBI REU students                                                   |
| 2019 | “Introduction to epidemic modelling”, SAMMS REU students                                             |
| 2019 | Mentor, MBI REU programme; Ohio 5 SURE students, College of Public Health, The Ohio State University |

### *Refereeing*

|                                                  |                                                     |
|--------------------------------------------------|-----------------------------------------------------|
| Electronic Journal of Probability                | PLOS Computational Biology                          |
| Journal of Applied Probability                   | Journal of Chemical Physics                         |
| Advances in Applied Probability                  | Journal of Mathematical Physics                     |
| Stochastic Processes and their Applications      | Journal of Mathematical Analysis and Applications   |
| Stochastics                                      | Journal of Mathematics in Industry                  |
| Statistics and Probability Letters               | Journal of Scientific Research                      |
| Methodology and Computing in Applied Probability | Afrika Matematika                                   |
| SIAM Journal on Applied Dynamical Systems        | Mathematics (MDPI)                                  |
| SIAM Multiscale Modeling & Simulation            | European Journal of Control                         |
| Biometrika                                       | IEEE Transactions on Automatic Control              |
| Mathematical Biosciences                         | IEEE Transactions on Networking                     |
| Bulletin of Mathematical Biology                 | IEEE Transactions on Control of Network Systems     |
| Journal of Mathematical Biology                  | IEEE Control Systems Letters                        |
| Journal of Theoretical Biology                   | IEEE Multi-Conference on Systems and Control (2015) |
| Theoretical Population Biology                   | American Control Conference (ACC)                   |
| Mathematical Biosciences and Engineering         |                                                     |
| PLOS One                                         |                                                     |

### *Reviewing and judging*

|            |                                                                                |
|------------|--------------------------------------------------------------------------------|
|            | Reviewer for AMS Mathematical Reviews (MathSciNet) and zbMATH Open             |
| 2021, 2023 | Judge, poster sessions, Annual Meeting of the Society for Mathematical Biology |
| 2021       | Judge, abstract submissions, Annual Meeting of the MIDAS Network               |

### *Memberships*

Royal Statistical Society (Applied Probability Section committee member)  
Society for Mathematical Biology (life member)

## **MEDIA**

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- **The Dynamical Systems Web (DSWeb)**: student feature, January 2021
- **OSC Research Report**: “COVID-19 projection models equip Ohioans”
- **OSC News** and **HPCwire**: “Pandemic modeling: COVID-19 projection models equip Ohioans”
- **IDI News**: “Far from trivial: a postdoctoral researcher’s path to a key position in the OSU/IDI COVID-19 modeling team”
- **STEAM Factory**: member spotlight

## LANGUAGES AND COMPUTING

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|           |                                                                                                            |
|-----------|------------------------------------------------------------------------------------------------------------|
| Languages | English (professional proficiency), Bengali (native), Hindi (working knowledge), German (intermediate, B1) |
| Computing | C, R, Python, SQL, Julia, MATLAB, SAS, L <sup>A</sup> T <sub>E</sub> X                                     |

## REFERENCES

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Available on request.